Linear Algebra and Its Applications

Section 1.7: Linear Independence

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

1.7 Linear independence 4

Practice problem 1.7.P1: Check every pair, not only the whole list 4

Practice problem 1.7.P2: Pairwise independence is weaker than independence 4

Practice problem 1.7.P3: Why testing a selected vector can miss dependence 4

Practice problem 1.7.P4: More vectors than coordinates 4

Exercise 1.7.1: Test the column equation 4

Exercise 1.7.2: Test the column equation 5

Exercise 1.7.3: Test the column equation 6

Exercise 1.7.4: Test the column equation 6

Exercise 1.7.5: Test the column equation 7

Exercise 1.7.6: Test the column equation 7

Exercise 1.7.7: Test the column equation 8

Exercise 1.7.8: Test the column equation . . . 9
Exercise 1.7.9: A span question can differ from a dependence question . . . 9
Exercise 1.7.10: The same distinction with another line . . . 10
Exercise 1.7.11: Locate the parameter value that removes a pivot . . . 10
Exercise 1.7.12: Locate the parameter value that removes a pivot . . . 10
Exercise 1.7.13: Locate the parameter value that removes a pivot . . . 11
Exercise 1.7.14: Locate the parameter value that removes a pivot . . . 11
Exercise 1.7.15: Four vectors in a two-coordinate space . . . 12
Exercise 1.7.16: Spot a scalar multiple . . . 12
Exercise 1.7.17: A zero vector is enough . . . 12
Exercise 1.7.18: Count the coordinates first . . . 12
Exercise 1.7.19: Two vectors that are not multiples . . . 13
Exercise 1.7.20: Another zero-vector relation . . . 13
Exercise 1.7.21: Four statements about dependence . . . 13
Exercise 1.7.22: Avoid converses that have not been proved . . . 13
Exercise 1.7.23: Every rank-deficient two-by-two echelon pattern . . . 13
Exercise 1.7.24: Three independent columns . . . 14
Exercise 1.7.25: Do not silently assume the first column is nonzero . . . 14
Exercise 1.7.26: A third column outside a plane . . . 14
Exercise 1.7.27: Count pivots for independence . . . 14
Exercise 1.7.28: Count pivots for spanning . . . 14
Exercise 1.7.29: Build dependent and independent examples . . . 15
Exercise 1.7.30: State the exact pivot condition . . . 15
Exercise 1.7.31: Read a relation from the columns . . . 15
Exercise 1.7.32: Read another relation without elimination . . . 15
Exercise 1.7.33: A relation already gives the answer . . . 15
Exercise 1.7.34: The overlooked zero-vector exception . . . 16
Exercise 1.7.35: A zero column yields a relation . . . 16
Exercise 1.7.36: The first two vectors might already be dependent . . . 16
Exercise 1.7.37: Dependence survives enlarging the list . . . 16
Exercise 1.7.38: Independence survives taking a sublist . . . 16
Exercise 1.7.39: Use the special target zero . . . 16
Exercise 1.7.40: No free variable means no second solution . . . 16
Exercise 1.7.41: Choose a maximal independent collection of original columns . . . 17

Exercise 1.7.42: Choose a maximal independent collection of original columns 17
Exercise 1.7.43: Recover an omitted column 18
Exercise 1.7.44: Why maximality forces every omitted column into the span 18

1.7 Linear independence

The main question. For vectors $v_1, \ldots, v_p$, write $c_1v_1+\cdots+c_pv_p = 0$. The list is *linearly independent* if its only solution is $c_1 = \cdots = c_p = 0$. It is dependent if a solution exists with at least one nonzero coefficient. Notice that the zero-coefficient solution always exists, so its existence proves nothing. Putting the vectors into the columns of A turns the question into $Ac = 0$. A pivot in every column means no free variable, hence independence. A free variable supplies a nontrivial dependence relation.

Practice problem 1.7.P1: Check every pair, not only the whole list

The four vectors are $u = (3, 2, -4)$, $v = (-6, 1, 7)$, $w = (0, -5, 2)$, and $z = (3, 7, -5)$. None is zero. For two nonzero vectors, dependence means one is a scalar multiple of the other. We can disprove this by looking just at their first two coordinates: the quantities $a_1b_2 - a_2b_1$ for $(u, v), (u, w), (u, z), (v, w), (v, z), (w, z)$ are respectively $15, -15, 15, 30, -45, 15$. None is zero. For example, if $v = tu$, then $3(1) - 2(-6)$ would have to be zero; it is 15.

Conclusion. All six pairs are linearly independent.

Practice problem 1.7.P2: Pairwise independence is weaker than independence

No. Checking pairs rules out a relation using only two vectors, but it does not rule out a relation using three or four. For instance, $e_1, e_2, e_1 + e_2$ are pairwise independent yet $e_1 + e_2 - (e_1 + e_2) = 0$ is a nontrivial relation among all three. The same logical distinction applies to the four given vectors.

Practice problem 1.7.P3: Why testing a selected vector can miss dependence

It is not a reliable test to ask only whether w is in the span of the other three. A dependence relation may have coefficient zero on w. For these data, solving with the actual coordinates gives $z = 3u + v$. Thus $3u + v - z = 0$ already establishes dependence without involving w. Moreover w is outside $\text{Span}\{u, v\}$: the equations $3a - 6b = 0$, $2a + b = -5$ force $b = -1$, $a = -2$, but then $-4a + 7b = 1 \neq 2$. So $w \notin \text{Span}\{u, v, z\}$ despite dependence of the whole list. Testing one chosen vector could therefore mislead us.

Practice problem 1.7.P4: More vectors than coordinates

There are four vectors in $\mathbb{R}^3$, so their 3×4 column matrix cannot have a pivot in all four columns. At least one variable is free in the homogeneous system, and the list is dependent. The relation $3u + v - z = 0$ from the preceding problem gives a direct check.

Exercise 1.7.1: Test the column equation

What we must show. Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 5 & 7 & 9 \\ 0 & 2 & 4 \\ 0 & -6 & -8 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{5}R_1$. Then

$$\begin{bmatrix} 1 & \frac{7}{5} & \frac{9}{5} \\ 0 & 2 & 4 \\ 0 & -6 & -8 \end{bmatrix}.$$

Use $R_2 \leftarrow \frac{1}{2}R_2, \quad R_1 \leftarrow R_1 - (\frac{7}{5})R_2, \quad R_3 \leftarrow R_3 - (-6)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Use $R_3 \leftarrow \frac{1}{4}R_3, \quad R_1 \leftarrow R_1 - (-1)R_3, \quad R_2 \leftarrow R_2 - (2)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.2: Test the column equation

What we must show. Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 0 & 0 & -1 \\ 2 & 0 & 3 \\ 3 & -8 & 1 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2, \quad R_1 \leftarrow \frac{1}{2}R_1, \quad R_3 \leftarrow R_3 - (3)R_1$. Then

$$\begin{bmatrix} 1 & 0 & \frac{3}{2} \\ 0 & 0 & -1 \\ 0 & -8 & -\frac{7}{2} \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3, \quad R_2 \leftarrow -\frac{1}{8}R_2$. Then

$$\begin{bmatrix} 1 & 0 & \frac{3}{2} \\ 0 & 1 & \frac{7}{16} \\ 0 & 0 & -1 \end{bmatrix}.$$

Use $R_3 \leftarrow -1R_3, \quad R_1 \leftarrow R_1 - (\frac{3}{2})R_3, \quad R_2 \leftarrow R_2 - (\frac{7}{16})R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.3: Test the column equation

What we must show. Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 2 & -4 \\ -3 & 6 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{2}R_1, \quad R_2 \leftarrow R_2 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

There is a free variable. For example, the nonzero coefficient vector $c = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ satisfies $Ac = 0$ (substitution checks every coordinate). This is a nontrivial relation among the columns.

Conclusion. The vectors are linearly dependent.

Exercise 1.7.4: Test the column equation

What we must show. Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} -1 & -3 \\ 3 & -9 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2, \quad R_1 \leftarrow \frac{1}{3}R_1, \quad R_2 \leftarrow R_2 - (-1)R_1$. Then

$$\begin{bmatrix} 1 & -3 \\ 0 & -6 \end{bmatrix}.$$

Use $R_2 \leftarrow -\frac{1}{6}R_2, \quad R_1 \leftarrow R_1 - (-3)R_2$. Then

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.5: Test the column equation

What we must show. Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 0 & -3 & 9 \\ 2 & 1 & -7 \\ -1 & 4 & -5 \\ 1 & -4 & -2 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_4$, $R_2 \leftarrow R_2 - (2)R_1$, $R_3 \leftarrow R_3 - (-1)R_1$. Then

$$\begin{bmatrix} 1 & -4 & -2 \\ 0 & 9 & -3 \\ 0 & 0 & -7 \\ 0 & -3 & 9 \end{bmatrix}.$$

Use $R_2 \leftarrow \frac{1}{9}R_2$, $R_1 \leftarrow R_1 - (-4)R_2$, $R_4 \leftarrow R_4 - (-3)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -\frac{10}{3} \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & -7 \\ 0 & 0 & 8 \end{bmatrix}.$$

Use $R_3 \leftrightarrow R_4$, $R_3 \leftarrow \frac{1}{8}R_3$, $R_1 \leftarrow R_1 - (-\frac{10}{3})R_3$, $R_2 \leftarrow R_2 - (-\frac{1}{3})R_3$, $R_4 \leftarrow R_4 - (-7)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.6: Test the column equation

What we must show. Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} -4 & -3 & 0 \\ 0 & -1 & 5 \\ 1 & 1 & -5 \\ 2 & 1 & -10 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_3$, $R_3 \leftarrow R_3 - (-4)R_1$, $R_4 \leftarrow R_4 - (2)R_1$. Then

$$\begin{bmatrix} 1 & 1 & -5 \\ 0 & -1 & 5 \\ 0 & 1 & -20 \\ 0 & -1 & 0 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_1 \leftarrow R_1 - (1)R_2$, $R_3 \leftarrow R_3 - (-1)R_2$, $R_4 \leftarrow R_4 - (-1)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 15 \\ 0 & 1 & -20 \\ 0 & 0 & -15 \\ 0 & 0 & -20 \end{bmatrix}.$$

Use $R_3 \leftarrow -\frac{1}{15}R_3$, $R_1 \leftarrow R_1 - (15)R_3$, $R_2 \leftarrow R_2 - (-20)R_3$, $R_4 \leftarrow R_4 - (-20)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.7: Test the column equation

What we must show. Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 1 & 4 & -3 & 0 \\ -2 & -7 & 5 & 1 \\ -4 & -5 & 7 & 5 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-2)R_1$, $R_3 \leftarrow R_3 - (-4)R_1$. Then

$$\begin{bmatrix} 1 & 4 & -3 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 11 & -5 & 5 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (4)R_2$, $R_3 \leftarrow R_3 - (11)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 6 & -6 \end{bmatrix}.$$

Use $R_3 \leftarrow \frac{1}{6}R_3$, $R_1 \leftarrow R_1 - (1)R_3$, $R_2 \leftarrow R_2 - (-1)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

There is a free variable. For example, the nonzero coefficient vector $c = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ satisfies $Ac = 0$ (substitution checks every coordinate). This is a nontrivial relation among the columns.

Conclusion. The vectors are linearly dependent.

Exercise 1.7.8: Test the column equation

What we must show. Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 1 & -2 & 3 & 2 \\ -2 & 4 & -6 & 2 \\ 0 & 1 & -1 & 3 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-2)R_1$. Then

$$\begin{bmatrix} 1 & -2 & 3 & 2 \\ 0 & 0 & 0 & 6 \\ 0 & 1 & -1 & 3 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3, \quad R_1 \leftarrow R_1 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 1 & 8 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix}.$$

Use $R_3 \leftarrow \frac{1}{6}R_3, \quad R_1 \leftarrow R_1 - (8)R_3, \quad R_2 \leftarrow R_2 - (3)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

There is a free variable. For example, the nonzero coefficient vector $c = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ satisfies $Ac = 0$ (substitution checks every coordinate). This is a nontrivial relation among the columns.

Conclusion. The vectors are linearly dependent.

Exercise 1.7.9: A span question can differ from a dependence question

(a) Reduce the span of the first two vectors.

Here $v_1 = (1, -3, 2)$ and $v_2 = (-3, 9, -6) = -3v_1$. Their span is the line of multiples of v_1. For $v_3 = (5, -7, h)$ to lie on that line, its first coordinate would force the multiplier to be 5, but its second would then have to be -15, not -7. No choice of h fixes that contradiction.

(b) Test the three-vector list instead.

The relation $3v_1 + v_2 + 0v_3 = 0$ is nontrivial for every h.

Conclusion. (a) No real h. (b) Every real h. The first two vectors are already dependent, whether or not the third lies in their span.

Exercise 1.7.10: The same distinction with another line

Here $v_1 = (1, -3, -5)$, $v_2 = (-3, 9, 15) = -3v_1$, and $v_3 = (2, -5, h)$.

(a) Compare coordinates of a proposed multiple.

If $v_3 = tv_1$, its first coordinate gives $t = 2$, but then the second coordinate would be -6, not -5. Thus v_3 is outside $\text{Span}\{v_1, v_2\}$ for every h.

(b) Exhibit a relation that does not use v_3.

Again $3v_1 + v_2 + 0v_3 = 0$, with coefficients not all zero.

Conclusion. (a) No h. (b) All $h \in \mathbb{R}$.

Exercise 1.7.11: Locate the parameter value that removes a pivot

Plan. Keep h symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about h.

$$A = \begin{bmatrix} 2 & 4 & -2 \\ -2 & -6 & 2 \\ 4 & 7 & h \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{2}R_1$, $\quad R_2 \leftarrow R_2 - (-2)R_1$, $\quad R_3 \leftarrow R_3 - (4)R_1$. Then

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & -1 & h+4 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $\quad R_2 \leftarrow -1R_2$, $\quad R_1 \leftarrow R_1 - (2)R_2$, $\quad R_3 \leftarrow R_3 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 2h+7 \\ 0 & 1 & -h-4 \\ 0 & 0 & -2h-8 \end{bmatrix}.$$

The first two pivot columns are present for all h. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $-2\,(h+4) = 0$.

Conclusion. The required value is $h = -4$. At other values all three coefficients are forced to zero, giving independence.

Exercise 1.7.12: Locate the parameter value that removes a pivot

Plan. Keep h symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about h.

$$A = \begin{bmatrix} 3 & -6 & 9 \\ -6 & 4 & h \\ 1 & -3 & 3 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_3$, $R_2 \leftarrow R_2 - (-6)R_1$, $R_3 \leftarrow R_3 - (3)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 3 \\ 0 & -14 & h+18 \\ 0 & 3 & 0 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_2 \leftarrow \frac{1}{3}R_2$, $R_1 \leftarrow R_1 - (-3)R_2$, $R_3 \leftarrow R_3 - (-14)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & h+18 \end{bmatrix}.$$

The first two pivot columns are present for all h. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $h + 18 = 0$.

Conclusion. The required value is $h = -18$. At other values all three coefficients are forced to zero, giving independence.

Exercise 1.7.13: Locate the parameter value that removes a pivot

Plan. Keep h symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about h.

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 5 & -9 & h \\ -3 & 6 & -9 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (5)R_1$, $R_3 \leftarrow R_3 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & h-15 \\ 0 & 0 & 0 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 2h-27 \\ 0 & 1 & h-15 \\ 0 & 0 & 0 \end{bmatrix}.$$

The first two pivot columns are present for all h. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $0 = 0$.

This condition holds identically. The reduced equations are solved nontrivially by $c_3 = 1$, $c_2 = 15 - h$, and $c_1 = 27 - 2h$.

Conclusion. The list is dependent for every real h.

Exercise 1.7.14: Locate the parameter value that removes a pivot

Plan. Keep h symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about h.

$$A = \begin{bmatrix} 1 & -3 & 2 \\ -2 & 7 & 1 \\ -4 & 6 & h \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-2)R_1, \quad R_3 \leftarrow R_3 - (-4)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 2 \\ 0 & 1 & 5 \\ 0 & -6 & h+8 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-3)R_2, \quad R_3 \leftarrow R_3 - (-6)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 17 \\ 0 & 1 & 5 \\ 0 & 0 & h+38 \end{bmatrix}.$$

The first two pivot columns are present for all h. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $h + 38 = 0$.

Conclusion. The required value is $h = -38$. At other values all three coefficients are forced to zero, giving independence.

Exercise 1.7.15: Four vectors in a two-coordinate space

The list has four vectors in $\mathbb{R}^2$. Its 2×4 column matrix has at most two pivots, leaving at least two free variables in $Ac = 0$.

Conclusion. Dependent. More than two vectors in $\mathbb{R}^2$ cannot be independent.

Exercise 1.7.16: Spot a scalar multiple

The second vector equals $-\frac{3}{2}$ times the first: $(-3, 6, -12) = -\frac{3}{2}(2, -4, 8)$. Thus $3v_1 + 2v_2 = 0$ is a relation with nonzero coefficients.

Conclusion. Dependent.

Exercise 1.7.17: A zero vector is enough

The second vector is zero. The coefficients $(0, 1, 0)$ produce $0v_1 + 1v_2 + 0v_3 = 0$ without being all zero.

Conclusion. Dependent. The values of the other vectors do not matter.

Exercise 1.7.18: Count the coordinates first

There are four vectors in $\mathbb{R}^2$, so a 2×4 column matrix cannot have four pivot columns. Its homogeneous system has a free variable.

Conclusion. Dependent. Row reduction is unnecessary.

Exercise 1.7.19: Two vectors that are not multiples

For $u = (-8, 12, -4)$ and $v = (2, -3, -1)$, the first coordinate of $v = tu$ would force $t = -1/4$. The third coordinate would then be 1, whereas v has third coordinate -1. Neither vector is zero, so the failure of scalar proportionality proves independence.

Conclusion. Independent.

Exercise 1.7.20: Another zero-vector relation

The third vector is zero. The coefficient list $(0, 0, 1)$ therefore solves the column equation nontrivially.

Conclusion. Dependent.

Exercise 1.7.21: Four statements about dependence

(a) **False.** Every homogeneous equation has the trivial solution. Independence requires it to be the *only* solution. The zero matrix has the trivial solution and many others.

(b) **False.** At least one vector is a combination of the others, not necessarily every vector. The list $e_1, 2e_1, e_2$ is dependent, but e_2 is not in the span of the first two vectors.

(c) **True.** Five columns in $\mathbb{R}^4$ give more variables than possible pivots, so the homogeneous system has a free variable.

(d) **True.** A nontrivial relation $ax + by + cz = 0$ must have $c \neq 0$; otherwise independence of x, y would give $a = b = 0$. Solving gives $z = -(a/c)x - (b/c)y$.

Exercise 1.7.22: Avoid converses that have not been proved

(a) **True.** If $w = au + bv$, then $au + bv - w = 0$ is a relation with coefficient -1 on w, hence a nontrivial relation.

(b) **False as written.** A plane need not pass through zero. The independent vectors e_1, e_2, e_3 all lie in the plane $x_1 + x_2 + x_3 = 1$. Vectors lying in a plane *through zero* are dependent when there are three of them, but the question does not impose that condition.

(c) **False.** In $\mathbb{R}^3$, the two vectors $e_1, 2e_1$ are dependent. Having few vectors makes independence possible, not automatic.

(d) **False.** The two vectors $e_1, 2e_1$ in $\mathbb{R}^3$ give a counterexample: dependence with only two vectors, fewer than three.

Exercise 1.7.23: Every rank-deficient two-by-two echelon pattern

Use p for a nonzero pivot and $*$ for an arbitrary entry. Dependent columns mean fewer than two pivots. The possible echelon patterns are

$$\begin{bmatrix} p & * \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & p \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

These exhaust rank one and rank zero. A second pivot would force independence.

Exercise 1.7.24: Three independent columns

Independence requires all three columns to have pivots. Since pivots move right as one moves down, their only possible positions are $(1,1),(2,2),(3,3)$. Thus the echelon pattern is

$$\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \end{bmatrix}, \qquad p,q,r \neq 0.$$

The arbitrary entries are above pivots; entries below them are zero.

Exercise 1.7.25: Do not silently assume the first column is nonzero

If $a_1 \neq 0$ and a_2 is not its multiple, the two columns are independent. But the hypothesis also permits $a_1 = 0$, in which case $a_2 \neq 0$ and there is only one pivot, in column two. Thus both patterns are possible:

$$\begin{bmatrix} p & * \\ 0 & q \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & p \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad p,q \neq 0.$$

For the second pattern, no nonzero vector is a multiple of the zero vector, so it really does satisfy the stated condition.

Exercise 1.7.26: A third column outside a plane

Independence of a_1, a_2 supplies two pivots. If the third column failed to add a pivot, it would be a combination of the first two, contrary to the assumption. Hence all three columns are pivot columns:

$$\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix}, \qquad p,q,r \neq 0.$$

Exercise 1.7.27: Count pivots for independence

A 6×4 matrix with independent columns must have **four** pivot columns: one for each variable in $Ax = 0$. Fewer would leave a free variable and hence a nonzero solution. Extra rows do not require extra pivots.

Exercise 1.7.28: Count pivots for spanning

For the columns of a 4×6 matrix to span $\mathbb{R}^4$, $Ax = b$ must be consistent for every four-coordinate b. This requires a pivot in every row, so there must be **four** pivot columns. Six are impossible because there are only four rows.

Exercise 1.7.29: Build dependent and independent examples

Take

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then $A(-2,1)^T = 0$ is a nontrivial solution. On the other hand, $Bx = 0$ has first two equations $x_1 = 0, x_2 = 0$, so its only solution is zero. Both matrices have the required size 3×2.

Exercise 1.7.30: State the exact pivot condition

(a) Fill the blank.

The number is $\boxed{n}$.

(b) Prove both directions.

There are n columns. If all n are pivot columns, $Ax = 0$ has no free variable, so its only solution is zero and the columns are independent. Conversely, if there are fewer than n pivot columns, some variable is free. Set one free variable to 1 and the others to 0, and solve for the pivot variables. This gives a nonzero homogeneous solution, contradicting independence. Thus independence forces n pivot columns.

Exercise 1.7.31: Read a relation from the columns

The given relation is $a_3 = a_1 + a_2$. Move all terms to one side: $a_1 + a_2 - a_3 = 0$. Therefore

$$x = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

is a nonzero solution of $Ax = 0$. No row reduction is needed, because matrix multiplication is the weighted sum of the columns.

Exercise 1.7.32: Read another relation without elimination

Here $a_1 - 3a_2 = a_3$, so $a_1 - 3a_2 - a_3 = 0$. Consequently $x = (1, -3, -1)^T$ is a nontrivial solution. The signs come from moving a_3 to the left, not from changing the columns themselves.

Exercise 1.7.33: A relation already gives the answer

True. The assumption $v_3 = 2v_1 + v_2$ gives $2v_1 + v_2 - v_3 + 0v_4 = 0$. The coefficients are not all zero, so the four-vector list is dependent regardless of v_4.

Exercise 1.7.34: The overlooked zero-vector exception

False. In $\mathbb{R}^4$, take $v_1 = 0$ and $v_2 = e_1$. The nonzero v_2 is not a scalar multiple of v_1, but $1v_1 + 0v_2 = 0$ is a nontrivial dependence relation. For two nonzero vectors the proposed test would be valid; the statement did not require both vectors to be nonzero.

Exercise 1.7.35: A zero column yields a relation

True. If $v_3 = 0$, the coefficient list $(0, 0, 1, 0, 0)$ gives a zero weighted sum and is not itself zero. Thus the list is dependent.

Exercise 1.7.36: The first two vectors might already be dependent

False. Let $v_1 = e_1$, $v_2 = 2e_1$, and $v_3 = e_2$ in $\mathbb{R}^3$. The third vector is outside the span of the first two, but $2v_1 - v_2 + 0v_3 = 0$ proves dependence. Adding a new direction cannot remove a relation that was already present.

Exercise 1.7.37: Dependence survives enlarging the list

True. A nontrivial relation among v_1, v_2, v_3 remains a nontrivial relation among v_1, v_2, v_3, v_4 after assigning coefficient zero to v_4. Therefore the larger list is dependent.

Exercise 1.7.38: Independence survives taking a sublist

True. Suppose $c_1v_1 + c_2v_2 + c_3v_3 = 0$. Then $c_1v_1 + c_2v_2 + c_3v_3 + 0v_4 = 0$. Independence of the four-vector list forces $c_1 = c_2 = c_3 = 0$. This is precisely the independence test for the three-vector sublist.

Exercise 1.7.39: Use the special target zero

What we must show. Prove column independence from the assumption of at most one solution for every b. Apply the assumption to $b = 0$. The zero vector is already one solution of $Ax = 0$, so "at most one" says there is no other solution. Thus every relation among the columns has all coefficients zero, which is the definition of independence. Existence for other b is not needed.

Exercise 1.7.40: No free variable means no second solution

There are n pivot columns and only n columns altogether, so there are no free variables. For a given b, reduction of $[A \mid b]$ may produce a contradiction, giving no solution; if it does not, every variable is a pivot variable with one determined value. Thus there can never be infinitely many solutions or two distinct solutions. Equivalently, if $Ax = Ay = b$, then $A(x - y) = 0$; the n pivots force $x - y = 0$, and hence $x = y$.

Exercise 1.7.41: Choose a maximal independent collection of original columns

What we must show. Retain as many original columns as possible while making $Bx = 0$ have only the zero solution. The source matrix is

$$A = \begin{bmatrix} 3 & -4 & 10 & 7 & -4 \\ -5 & -3 & -7 & -11 & 15 \\ 4 & 3 & 5 & 2 & 1 \\ 8 & -7 & 23 & 4 & 15 \end{bmatrix}.$$

Exact row reduction gives

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its pivot columns are 1, 2, 4. Take these columns from the *original* matrix, not from its reduced matrix:

$$B = \begin{bmatrix} 3 & -4 & 7 \\ -5 & -3 & -11 \\ 4 & 3 & 2 \\ 8 & -7 & 4 \end{bmatrix}.$$

Reducing B gives

$$\operatorname{rref}(B) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $Bx = 0$ forces every one of its 3 variables to be zero. No larger independent subcollection is possible, because the reduced A has only 3 pivot positions; every nonpivot column is a combination of the pivot columns. The next related exercise exhibits such a combination explicitly.

Exercise 1.7.42: Choose a maximal independent collection of original columns

What we must show. Retain as many original columns as possible while making $Bx = 0$ have only the zero solution. The source matrix is

$$A = \begin{bmatrix} 12 & 10 & -6 & 8 & 4 & -14 \\ -7 & -6 & 4 & -5 & -7 & 9 \\ 9 & 9 & -9 & 9 & 9 & -18 \\ -4 & -3 & -1 & 0 & -8 & 1 \\ 8 & 7 & -5 & 6 & 1 & -11 \end{bmatrix}.$$

Exact row reduction gives

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 & 0 & -2 \\ 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its pivot columns are 1, 2, 3, 5. Take these columns from the *original* matrix, not from its reduced matrix:

$$B = \begin{bmatrix} 12 & 10 & -6 & 4 \\ -7 & -6 & 4 & -7 \\ 9 & 9 & -9 & 9 \\ -4 & -3 & -1 & -8 \\ 8 & 7 & -5 & 1 \end{bmatrix}.$$

Reducing B gives

$$\operatorname{rref}(B) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus $Bx = 0$ forces every one of its 4 variables to be zero. No larger independent subcollection is possible, because the reduced A has only 4 pivot positions; every nonpivot column is a combination of the pivot columns. The next related exercise exhibits such a combination explicitly.

Exercise 1.7.43: Recover an omitted column

For the choice $B = [a_1\ a_2\ a_4]$ in Exercise 41, choose $v = a_3 = (10, -7, 5, 23)^T$. The reduced column relation from that exercise is $a_3 = 2a_1 - a_2$. Directly,

$$2\begin{bmatrix} 3 \\ -5 \\ 4 \\ 8 \end{bmatrix} - \begin{bmatrix} -4 \\ -3 \\ 3 \\ -7 \end{bmatrix} = \begin{bmatrix} 10 \\ -7 \\ 5 \\ 23 \end{bmatrix}.$$

So $Bc = v$ is solved by $c = (2, -1, 0)^T$.

Conclusion. v is in the span of the columns of B.

Exercise 1.7.44: Why maximality forces every omitted column into the span

For Exercise 42, $B = [a_1\ a_2\ a_3\ a_5]$. Choose the omitted column $v = a_4$. Its coordinates in the retained columns are $(1, -1, -1, 0)^T$, because

$$a_1 - a_2 - a_3 = \begin{bmatrix} 12 \\ -7 \\ 9 \\ -4 \\ 8 \end{bmatrix} - \begin{bmatrix} 10 \\ -6 \\ 9 \\ -3 \\ 7 \end{bmatrix} - \begin{bmatrix} -6 \\ 4 \\ -9 \\ -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 9 \\ 0 \\ 6 \end{bmatrix} = a_4.$$

Thus $v \in \operatorname{Col} B$. In general, if an omitted column were outside this span, adjoining it to the independent columns of B would produce a larger independent list: a relation with nonzero coefficient on the new column would express it in the old span, and a relation with zero coefficient would contradict independence of the old list. That contradicts the requirement that B use as many columns as possible.

Conclusion. Every omitted column lies in the span of a maximal independent subcollection, not only the particular column checked here.